

Multiple Temporal Fusion based Weakly-supervised Pre-training Techniques for Video Categorization

Xiaochen Cai

4Paradigm Inc.

Beijing, China

caixiaochen@4paradigm.com

Hengxing Cai

4Paradigm Inc.

Beijing, China

caihengxing@4paradigm.com

Boqing Zhu

National University of Defense

Technology

Changsha, China

zhubq@gmail.com

Kele Xu*

National University of Defense

Technology

Changsha, China

xukelele@163.com

Weiwei Tu

4Paradigm Inc.

Beijing, China

tuweiwei@4paradigm.com

Dawei Feng

National University of Defense

Technology

Changsha, China

ABSTRACT

In this paper, we present our solution of the ACM Multimedia 2022 pre-training for video understanding challenge. First, we pre-train the models on large-scale weakly-supervised video datasets with different temporal resolutions, then fine-tune the model for downstream application. Quantitative comparisons are conducted to evaluate the performance of different networks at multiple temporal resolutions. Moreover, we fusion different pre-trained models through weighted averaging. We achieve an accuracy of 62.39% in the testing set, which ranked as the first place in the video categorization track of this challenge.

CCS CONCEPTS

• **Computing methodologies** → **Computer vision representations**; *Neural networks*; *Computer vision tasks*.

KEYWORDS

Representation Learning, Video Understanding, Pre-training, Video Transformer, Temporal Resolution

ACM Reference Format:

Xiaochen Cai, Hengxing Cai, Boqing Zhu, Kele Xu, Weiwei Tu, and Dawei Feng. 2022. Multiple Temporal Fusion based Weakly-supervised Pre-training Techniques for Video Categorization. In *Proceedings of the 30th ACM International Conference on Multimedia (MM '22)*, October 10–14, 2022, Lisboa, Portugal. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3503161.3551585>

*Corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

MM '22, October 10–14, 2022, Lisboa, Portugal

© 2022 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 978-1-4503-9203-7/22/10...\$15.00

<https://doi.org/10.1145/3503161.3551585>

1 INTRODUCTION

The quality of the video representation is directly related to the performance of the video understanding task [11, 20], including the video classification and categorization. This paper focuses on pre-training approaches for the video understanding, endeavoring to learn generic video representations which can facilitate the performance of downstream tasks.

Recent studies [1, 5, 10, 14] have evidently demonstrated that the pre-training on large-scale image datasets followed by fine-tuning on target datasets may significantly boost the performance in image-related downstream tasks, for example, image classification [9], semantic segmentation and object detection [12]. One would certainly wonder whether these successes would carry over from images [15, 19] to the videos. On the other hand, the frame-level annotation for large-scale videos is usually impracticable, while the weakly-supervised videos could be collected relatively easily through methods such as web crawling. Therefore, the pre-training on weakly-supervised videos can be more realistic in practical settings. However, the labels of the weakly-supervised videos only indicate the presence or absence of the objects or actions in the entire video clip, without any onset and offset time annotations or the sequence of the objects or actions. Compared to pretraining techniques for the images, the pre-training for weakly-supervised video presents more challenges.

Firstly, the label of an image can be seen as frame-level annotation in the video, because a typical image is a particular moment in time. These web images usually suffer from the noisy label problem (i.e. missing or incorrect object label). In the weakly-supervised pre-training for the video, the noise problem is more serious. In addition to label noise, there is also temporal noise due to the lack of localization of objects or actions. For example, a video is annotated with *playing basketball*, yet only some frames are people playing basketball and other frames are the crowd watching the game. This makes the pre-training task more challenging under the weakly-supervised settings.

Secondly, with the involvement of the temporal dimension, different temporal resolutions may result in the different performances of the pre-trained networks. The temporal resolution is reflected by the sampling rate of the video frames which input to the network.

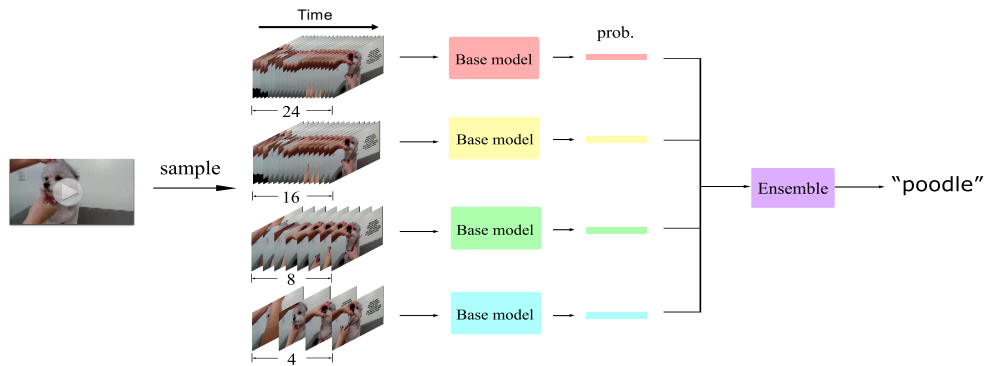


Figure 1: Fusion with Multiple Temporal Resolutions.

Specifically, for the same video clip, a high temporal resolution input will contain more frames, while they might be more redundant and noisy. The batch size needs to be smaller due to limited storage. In contrast, a low temporal resolution input may lose some relevant frames, but more samples can be accommodated in a mini-batch. Currently, it is not clear which temporal resolution can provide superior performance for pre-training of video representation learning.

Thirdly, frame-based models [8–10, 17], in which the input frames are regarded as independent samples, have already achieved promising results in video action recognition. Recently, various spatiotemporal models [3, 18] have been proposed, in which the input frames sequence were also considered. Whether could these spatial-temporal models boost the performance of video categorization task? How much additional computation and parameters are needed?

In this work, we aim to address aforementioned challenges by designing a novel pre-training technique for video understanding. First, we pre-train the models on the ImageNet dataset and three large-scale weakly-supervised video datasets, including Kinetics-400, Kinetics-600, and Something-Something-V2 datasets. And then pre-trained models are fine-tuned for downstream application, which is a 240-ways video categorization task used in the competition. Next, we evaluate the performance of different networks at different temporal resolutions. Subsequently, we compare the performance of some recently proposed frame-based models and spatiotemporal models, and we also investigate their computational and parametric costs. Finally, we ensemble seven pre-trained models by weighted averaging to obtain our final system. In terms of results, we achieve an accuracy of 62.39% in the testing set, which ranked as the first place in the video categorization track of the ACM Multimedia 2022 pre-training for video understanding challenge [21, 24].

The remainder of this paper is organized as follows: Section 2 presents the methodology details while the experiments and results of the challenge are given in Section 3. Section 4 draws the conclusion.

2 METHODOLOGY

In this section, we firstly introduce various network architectures employed for the challenge, including frame-based networks and

spatiotemporal networks. Then, we propose a fusion method based on multiple temporal resolutions.

2.1 Frame-Based Network Architectures

For the frame-based models, the backbone architectures are adapted from those for images by extending the modeling through the temporal axis. For example, the 3D convolution is a direct extension of 2D convolution for joint spatial and temporal modeling at the operator level. For the 2D Transformer architectures, the results are calculated independently for each frame, and then produce the result for the video clip by a probability voting. Specifically, three kinds of frame-based architectures are employed.

Temporal Segment Network: The Temporal Segment Networks (TSN) [22, 23] is a framework for video-based human action recognition which is proposed by Wang et al. TSN effectively models long-range temporal dynamics by learning from multiple segments of one video in an end-to-end manner. To be specific, TSN is composed of spatial stream ConvNets and temporal stream ConvNets. Instead of processing a single frame or frame stack, the temporal segment networks operate on a series of short clips sampled sparsely from the entire video. Each clip in this sequence will generate a preliminary prediction of the category of its own actions. Then a consensus is reached among the clips as a video level prediction.

Swin Transformer: Swin Transformer [17] is a new Transformer framework. Based on the ViT model, it introduces a hierarchical Transformer with shifted windows self-attention, which can learn the information within and across windows. Meanwhile, this hierarchical networks has the flexibility to model at various scales, which can significantly reduce the computation complexity compared with ViT.

BEiT: Motivated by BERT [8] developed in the natural language processing area, [2] proposed a self-supervised vision representation model named BEiT. The main idea of BEiT is a masked image modeling strategy to pre-train vision Transformers. In more detail, they first tokenize all the input patches into vision tokens and then randomly mask some of them. The pre-training objective is to recover the masked tokens which is similar to masked language model in BERT. Experimental results shows that pre-trained model has a

better representation ability and outperforms other counterparts in downstream task.

2.2 Spatiotemporal Network Architectures

For the spatiotemporal networks, the input frames are extended from independent images to the spatiotemporal 3D volume. Then, a sequence of patches is extracted from this volume. Each patch is linearly mapped into an embedding with positional information. This makes it possible to interpret the resulting sequence of vectors as token embeddings which can be fed to a Transformer encoder. In our approach, we tried two kinds of pure attention-based architectures named TimeSformer and Video Swin Transformer.

TimeSformer: TimeSformer is proposed in [3], which divides the videos into subsets of disjoint image blocks sequences, and use an unique self-attention method which performs self-attention on tokens at the same position across different frames or tokens in the same frame to avoid complex computation among all image block sequences. Different from models described above, TimeSformer is a 3D model that views all input frames together and generates a single prediction of the category.

Video Swin Transformer: Video Swin Transformer [18] is a variant of Swin Transformer [17] which is designed for video. By adapting the Swin Transformer from 2D to 3D, it introduces an inductive bias of locality in video Transformers, reducing computation complexity while maintaining similar accuracy compared to previous Transformer-based approaches. This approach achieves state-of-the-art accuracy on a broad range of video recognition benchmarks, including action recognition and temporal modeling.

2.3 Ensemble on Multiple Temporal Resolutions

The main idea of Bootstrap aggregating (Bagging) algorithm [4] is to construct different subsets of training sets by random sampling and then train different classifiers respectively. The final output is obtained by model averaging using the average value, majority votes or other methods. The model learns different knowledge from different data subsets and produces different errors in the test set, thus reducing the average generalization error of the models. Inspired by this, we suppose that for video streaming data, we can sample different data subsets from time domain instead of sampling from training sets. By this means, each base model can use all the videos in training sets and get different data subsets at the same time. Each base model can use all the training sets to ensure that the model has a low deviation. At the same time, different time-domain samples allow us to get different models, so as to reduce the variance through model averaging.

3 EXPERIMENTS

3.1 Datasets

Four widely used datasets are employed for the pre-training task: including the ImageNet-22k [7], Kinetics-400 [16], Kinetics-600 [6], and Something-Something-V2 [13]. For ImageNet-22k, the datasets consists of 14.2 million images and 22k classes, which is used to train the frame-based models. Kinetics-400 consists of $\sim 260k$ video clips which cover 400 human action classes. Kinetics-600 is an extension of Kinetics-400 that contains $\sim 400k$ videos from 600 human action classes. Each video clip in Kinetics datasets lasts around 10

seconds. Something-Something-V2 dataset is a collection of $\sim 220k$ weakly labeled video clips of 174 human-object interactions. Unfortunately, YOYO-3M, the official pre-training dataset for this challenge is not used in our pre-training, due to the heavy label noise. YOYO-Downstream dataset is used for the video categorization downstream evaluation. This dataset consists of a training set of 70,173 videos, a validation set of 16,439 videos and a test set of 16,554 videos. Video are divided into 240 pre-defined categories for objects (e.g. *Aircraft*, *Pizza* and *Football*) and human actions (e.g. *Waggle*, *High jump*, and *Riding*). Each video clip lasts for 2 seconds. Since the label of the test set is not public, all our experiments are based on the verification set.

3.2 Implementations details

All experiments are implemented based on mmaction2 framework¹ and we refer to the respective model codes and re-implement them in mmaction2. During the training phase, we sampled one frame every *interval* frames and sampling *total_frames* frames in total, and applied two augmentation operations: multi-scale crop and flipping. Mean and standard deviation for each channel is adopted from the ImageNet datasets (mean=[123.675, 116.28, 103.53], std=[58.395, 57.12, 57.375]). We employ an AdamW optimizer for 50 epochs unless specified, using a cosine annealing learning rate scheduler and 2-epoch linear warm-up. The learning rate is initialized as $3e-4$ and the weight decay is set as 0.05. Because the GPU memory consumption of models varies greatly and we do not have sufficient computing resources in practical settings, we use different batch-size which will be specified later. During the validation and test phase, all settings are the same except that we apply no augmentation operations.

For TSN-ResNet50 model, we pre-train the parameters on Kinetics400 dataset and then fine-tune it on downstream dataset for 50 epochs. The batch-size is set to 32. Since TSN is a frame-Based model which generate independent prediction results for each input frames, we take the average of the prediction for each frame as the model output. For all frame-Based model we take the same strategy. For Swin Transformer model, the official Swin-L model is used. we pre-train the model on ImageNet22k dataset and then fine-tune it on downstream dataset for 20 epochs, which is the best epoch on validation set. The batch-size is set to 16. For BEiT, we use the pre-train checkpoints offered by original author which is pre-trained on ImageNet22k by masked image modeling task and then fine-tune it on downstream dataset for 20 epochs. The batch-size is set as 8.

For TimeSformer model, the divST attention is used for our model. We pre-train it on Kinetics600 dataset and then fine-tune it on downstream dataset for 50 epochs. The batch-size is set to 8. As for Video Swin Transformer, since it surpasses all other models with the same configuration, we train it with different method and include all of them in our final submission. We pre-train them on Kinetics600 or Something-Something V2 dataset and then fine-tune it on downstream dataset for 50 epochs. The batch-size is set to 16.

3.3 Results

We compare the performance of frame-based models and spatiotemporal models mentioned above, and we also investigate their computational and parametric costs. The performance and computational

¹<https://github.com/open-mmlab/mmdetection>

Table 1: Performance of our submission system. The final system is an weighted fusion of 7 models with different temporal resolutions.

Backbone	Weight	Pre-train dataset	Sampling Rate	Single-model Accuracy	Ensemble Accuracy	
TSN-ResNet50	0.025	Kinetics-400	8/6	0.5214	0.6254	0.6239
TimeSformer	0.025	Kinetics-600	8/6	0.5440		
Video Swin Transformer	0.45	Kinetics-600	8/6	0.6140		
Video Swin Transformer	0.3	Kinetics-600	24/2	0.6064		
Video Swin Transformer	0.1	Something-Something-V2	24/2	0.5612		
Swin Transformer	0.05	ImageNet-22k	8/6	0.5823		
BEiT	0.05	ImageNet-22k	8/6	0.5758		

Table 2: Quantitative comparison by using different backbones.

Backbone	Accuracy	#params	FLOPs
TSN-ResNet50	0.5214	24M	33.0G
TimeSformer	0.5440	121.4M	189.0G
BEiT-Large	0.5758	202.6M	358.3G
Swin Transformer-Large	0.5823	195.4M	272.3G
Video Swin Transformer	0.6140	87.9M	60.6G

Table 3: Performance comparison with different pre-train datasets.

Backbone	Pre-train dataset	Accuracy
TimeSformer	Something-Something-V2	0.4716
TimeSformer	Kinetics-600	0.5440
Video Swin Transformer	Something-Something-V2	0.5612
Video Swin Transformer	Kinetics-600	0.6064

costs of different network structures are summarized in Table 2. This part of the experiment is carried out by using a sampling rate of $total_frames = 8$ and $interval = 6$. Video Swin Transformer reaches the highest single-model top-1 accuracy of 61.4% with second lowest complexity. So we take the Video Swin Transformer as the backbone for the follow-up experiments.

To analyse the effects of pre-train dataset, we compare the performance of two widely used video datasets: Kinetics600 and Something-Something-V2. We use TimeSformer and Video Swin Transformer as video classifiers to train a classification task on the pre-training dataset, 174 categories for Something-Something-V2 and 600 categories for Kinetics600, and then fine-tune and evaluate their performance on downstream task. We also tried to perform pre-training on the official pre-training dataset, but there is much noise in the official pre-training dataset and it perform significantly worse than the two datasets mentioned above. Therefore, we did not include the official pre-training dataset in the final results.

In order to verify the effectiveness of the fusion on multiple temporary resolutions and find the best temporal sample method as well, we trained our backbones at different sampling rates and compared them with ensemble using Bagging method. The results are shown in Table 4. Ensemble on multiple temporary resolutions strategy enables each base model to achieve better performance, and the performance after ensemble is significantly better than both the method of randomly sampling the training set to construct subsets. Ensemble on multiple temporary resolutions may be used as a new method to improve the accuracy of single model. On the other hand,

Table 4: Performance of ensemble on multiple temporal resolutions.

	Accuracy
subset 0	0.5789
subset 1	0.5805
subset 2	0.5791
subset 3	0.5802
Ensemble	0.6073
SR 4/12	0.6033
SR 8/6	0.6140
SR 16/3	0.5986
SR 24/2	0.6064
Ensemble	0.6193

Our experiments shows that sampling with $total_frames = 8$ and $interval = 6$ performs the best, but this result may not be universal and vary with different scenarios. High temporal resolution may generate redundancy and thus increase the risk of over fitting while low temporal resolution may lose some relevant frames. It is not clear what temporal resolution is better for video understanding yet.

We ensemble results from multiple models on the validation set. To be specific, each model generate categorization probabilities independently and we weight these probabilities to get a final results. We then perform a grid search to get the optimal combination coefficients. It is found that although the single model accuracy of TSN, TimeSformer, etc. fail to surpass Video Swin Transformer based method, these models can still contribute to the ensemble results. As shown in the Table 1, we get the optimal validation result of 62.54% and test result of 62.39%.

4 CONCLUSION

In this challenge paper, we present our solution for the ACM Multimedia 2022 pre-training for video understanding challenge, which ranked as the first place in the video categorization track. Here, we follow the pre-training and fine-tuning paradigm. Quantitative comparisons are conducted to demonstrate the effectiveness of the fusion with different temporal resolutions.

ACKNOWLEDGMENTS

This work is supported by the major Science and Technology Innovation 2030 “New Generation Artificial Intelligence” project 2021ZD0112904.

REFERENCES

- [1] Mykhaylo Andriluka, Umar Iqbal, Eldar Insafutdinov, Leonid Pishchulin, Anton Milan, Juergen Gall, and Bernt Schiele. 2018. Posetrack: A benchmark for human pose estimation and tracking. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 5167–5176.
- [2] Hangbo Bao, Li Dong, and Furu Wei. 2021. Beit: Bert pre-training of image transformers. *arXiv preprint arXiv:2106.08254* (2021).
- [3] Gedas Bertasius, Heng Wang, and Lorenzo Torresani. 2021. Is space-time attention all you need for video understanding?. In *ICML*, Vol. 2. 4.
- [4] Leo Breiman. 1996. Bagging predictors. *Machine learning* 24, 2 (1996), 123–140.
- [5] Zhe Cao, Tomas Simon, Shih-En Wei, and Yaser Sheikh. 2017. Realtime multi-person 2d pose estimation using part affinity fields. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 7291–7299.
- [6] Joao Carreira, Eric Noland, Andras Banki-Horvath, Chloe Hillier, and Andrew Zisserman. 2018. A short note about kinetics-600. *arXiv preprint arXiv:1808.01340* (2018).
- [7] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. 2009. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*. Ieee, 248–255.
- [8] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805* (2018).
- [9] Jeff Donahue, Yangqing Jia, Oriol Vinyals, Judy Hoffman, Ning Zhang, Eric Tzeng, and Trevor Darrell. 2014. Decaf: A deep convolutional activation feature for generic visual recognition. In *International conference on machine learning*. PMLR, 647–655.
- [10] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xi-aohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. 2020. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929* (2020).
- [11] Deepti Ghadiyaram, Du Tran, and Dhruv Mahajan. 2019. Large-scale weakly-supervised pre-training for video action recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 12046–12055.
- [12] Ross Girshick, Jeff Donahue, Trevor Darrell, and Jitendra Malik. 2014. Rich feature hierarchies for accurate object detection and semantic segmentation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 580–587.
- [13] Raghav Goyal, Samira Ebrahimi Kahou, Vincent Michalski, Joanna Materzynska, Susanne Westphal, Heuna Kim, Valentin Haenel, Ingo Fruend, Peter Yianilos, Moritz Mueller-Freitag, et al. 2017. The "something something" video database for learning and evaluating visual common sense. In *Proceedings of the IEEE international conference on computer vision*. 5842–5850.
- [14] Minyoung Huh, Pulkit Agrawal, and Alexei A Efros. 2016. What makes ImageNet good for transfer learning? *arXiv preprint arXiv:1608.08614* (2016).
- [15] Armand Joulin, Laurens van der Maaten, Allan Jabri, and Nicolas Vasilache. 2016. Learning visual features from large weakly supervised data. In *European Conference on Computer Vision*. Springer, 67–84.
- [16] Will Kay, Joao Carreira, Karen Simonyan, Brian Zhang, Chloe Hillier, Sudheendra Vijayanarasimhan, Fabio Viola, Tim Green, Trevor Back, Paul Natsev, et al. 2017. The kinetics human action video dataset. *arXiv preprint arXiv:1705.06950* (2017).
- [17] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. 2021. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 10012–10022.
- [18] Ze Liu, Jia Ning, Yue Cao, Yixuan Wei, Zheng Zhang, Stephen Lin, and Han Hu. 2022. Video swin transformer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 3202–3211.
- [19] Dhruv Mahajan, Ross Girshick, Vignesh Ramanathan, Kaiming He, Manohar Paluri, Yixuan Li, Ashwin Bharambe, and Laurens Van Der Maaten. 2018. Exploring the limits of weakly supervised pretraining. In *Proceedings of the European conference on computer vision (ECCV)*. 181–196.
- [20] Joe Yue-Hei Ng, Jonghyun Choi, Jan Neumann, and Larry S Davis. 2018. Actionflownet: Learning motion representation for action recognition. In *2018 IEEE Winter Conference on Applications of Computer Vision (WACV)*. IEEE, 1616–1624.
- [21] Yingwei Pan, Yehao Li, Jianjie Luo, Jun Xu, Ting Yao, and Tao Mei. 2020. Auto-captions on GIF: A Large-scale Video-sentence Dataset for Vision-language Pre-training. *arXiv preprint arXiv:2007.02375* (2020).
- [22] Limin Wang, Yuanjun Xiong, Zhe Wang, Yu Qiao, Dahua Lin, Xiaoou Tang, and Luc Van Gool. 2016. Temporal segment networks: Towards good practices for deep action recognition. In *European conference on computer vision*. Springer, 20–36.
- [23] Limin Wang, Yuanjun Xiong, Zhe Wang, Yu Qiao, Dahua Lin, Xiaoou Tang, and Luc Van Gool. 2018. Temporal segment networks for action recognition in videos. *IEEE transactions on pattern analysis and machine intelligence* 41, 11 (2018), 2740–2755.
- [24] Jun Xu, Tao Mei, Ting Yao, and Yong Rui. 2016. MSR-VTT: A Large Video Description Dataset for Bridging Video and Language. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.